

# THE SIGNALRUPTURE STRUCTURAL GRAMMAR

## A Unified Cross-Domain Empirical Synthesis of Healthcare, Electricity, and Households

SignalRupture (SR)

Post-Gauntlet Empirical Synthesis — August 2026

### Abstract

This manuscript combines the first three empirical probes of SignalRupture (SR) after its reformulation as a developing structural metatheory and cross-domain structural grammar. The synthesis tests whether a fixed relational architecture can be translated into three heterogeneous systems without changing its logical roles after outcomes are known. The frozen architecture distinguishes effective requirement (Q), durable capacity (D), sustainable flexibility (C\_S), extraordinary compensation (C\_E), and observable function (F). Canadian healthcare provides the strongest direct empirical case: regular staffing lost share while overtime and purchased staffing expanded substantially from 2019–2020 to 2023–2024, especially in rural and remote hospitals, while large-scale service delivery continued and later access indicators remained under pressure. North American electricity provides a hostile engineered-system test: the same relational positions can be specified using reliability-conditioned demand, dependable capacity, sustainable reserve/flexibility, exceptional substitution, and reliable delivery, but current evidence does not yet demonstrate a completed extraordinary-compensation-to-discontinuity sequence. Canadian households provide a heterogeneous social test: housing and food-security pressures identify exposed populations, but aggregate debt cannot validly stand in for extraordinary compensation; longitudinal household-level data are required to observe gap, coping, continuity, and later outcome in sequence.

The combined result is therefore neither full confirmation nor failure. Across three distinct domains, the grammatical roles Q, D, C\_S/C\_E, and F remain operationally stable. This establishes three-domain structural translation and supports the limited proposition that observable continuity is insufficient to infer durable structural sufficiency. Direct evidence for Compensated Continuity is strongest in healthcare;

electricity supports cross-domain compatibility and a measurable restoration race; households support operational translation while rejecting several invalid proxies. The stronger metatheoretical claim—that compensation state and trajectory provide incremental prospective information about later function beyond established domain models—remains unvalidated. SR has therefore crossed the operational-invariance threshold but not the predictive-validation threshold.

Keywords: SignalRupture; structural grammar; metatheory; cross-domain validation; compensated continuity; healthcare; electricity; households; empirical synthesis.

## 1. Purpose of the Unified Test

The first three empirical papers were deliberately separated so that the theory could not be modified in one domain to fit the others. This synthesis now combines them at the level of invariant relational structure rather than raw units. Hospital overtime hours, megawatts of reserve, and household savings or borrowing are not commensurable quantities. What can be compared is the role each measure occupies in the frozen grammar.

$$Q \leftrightarrow D \leftrightarrow C\_S / C\_E \leftrightarrow F$$

The empirical question is whether these roles survive translation, whether any domain supplies observed evidence for the predicted relations, and whether the combined evidence justifies a stronger metatheoretical claim.

## 2. Frozen Cross-Domain Grammar

| Symbol | Canonical role          | Cross-domain interpretation                                              |
|--------|-------------------------|--------------------------------------------------------------------------|
| Q      | Effective requirement   | What the system must supply to maintain the declared function or target. |
| D      | Durable capacity        | What the system can sustainably supply under ordinary operation.         |
| C_S    | Sustainable flexibility | Renewable, planned, or ordinary buffering.                               |

|     |                            |                                                                                  |
|-----|----------------------------|----------------------------------------------------------------------------------|
| C_E | Extraordinary compensation | Exceptional, costly, temporary, constrained, or non-renewable bridging capacity. |
| F   | Observable function        | The service, continuity, reliability, or social reproduction actually delivered. |

$$G\_C = Q - D$$

Compensated Continuity:  $G\_C > 0 \wedge C\_E \uparrow \wedge F \approx \text{maintained}$

Compensatory Inversion:  $\Delta C\_E \leq 0 \wedge G\_C > 0 \wedge \Delta F < 0$

### 3. Domain I — Canadian Healthcare

Healthcare supplies the strongest empirical foothold because compensation is directly observed. CIHI's staffing series shows a material shift in hospital staffing composition between 2019–2020 and 2023–2024. Nationally in the available series, purchased staffing hours rose 126% and overtime hours rose 100%. In rural and remote hospitals, purchased staffing rose 261%, compared with 98% in urban hospitals. The share of regular hours fell while overtime and purchased staffing gained share.

At the same time, hospitals continued delivering large volumes of care. Later CIHI emergency-department evidence records 16.1 million ED visits in 2024–2025 while wait pressure remained elevated and 7.7% of visits ended with patients leaving before physician assessment. The empirical configuration is therefore not system cessation. It is continued large-scale function with a materially altered staffing substrate and access pressure.

Staffing substrate shifts toward compensation while F\_volume remains substantial

This is the clearest observed example of Compensated Continuity in the three-paper program. The result remains descriptive rather than causal because a fully measured Q–D gap, sustainable overtime baseline, and matched staffing-to-outcome panel have not yet been estimated.

### 4. Domain II — North American Electricity

Electricity is the hostile non-human test. NERC's 2025 Long-Term Reliability Assessment projects more than 224 GW of additional summer peak demand over the ten-year horizon, with the forecast substantially higher than in the prior assessment. This makes Q a rapidly moving requirement and creates an observable race between realized demand and dependable resource additions.

The crucial correction produced by the electricity test is that normal reserve and flexibility are not evidence of structural deterioration. They belong to C\_S when they are planned, renewable, and embedded in normal adequacy criteria. C\_E begins only with exceptional actions outside ordinary planning assumptions. This distinction prevents SR from pathologizing resilience.

$D + C_S \geq Q \rightarrow \text{ordinary resilient operation}$

$Q > D + C_S$  and exceptional substitution preserves F  $\rightarrow$  candidate  
Compensated Continuity

The electricity domain therefore supports the grammar at the level of operational translation and measurable timing, but it does not yet provide a completed continental sequence showing persistent C\_E followed by constraint and discontinuity. Its current status is cross-domain structural compatibility, not confirmation of Compensatory Inversion.

## **5. Domain III — Canadian Households**

Households provide the most difficult social test because compensation is heterogeneous and partly hidden. The frozen translation defines Q as the predeclared cost of essential social reproduction, D as stable recurring after-tax resources, C\_S as renewable household buffering, C\_E as identifiable exceptional gap-bridging mechanisms, and F as continued housing, food security, utilities, essential care, solvency, and minimum social reproduction.

Canadian housing evidence identifies materially greater exposure among renters than owners, and food-security evidence identifies substantial distributional strain. However, the household study rejects the shortcut of treating aggregate debt as C\_E. Debt may finance assets, smooth temporary consumption, or coexist with substantial liquidity. It

becomes extraordinary compensation only when the data show that it is being used to bridge an identified essential-resource gap and is non-renewable or structurally persistent.

$$\text{Debt} \neq \text{C\_E}$$

The household grammar therefore survives operationally, but the complete compensated-continuity sequence remains unconfirmed without longitudinal microdata linking the same household's gap, coping response, temporary continuity, and later restoration or discontinuity.

## 6. The Cross-Domain Invariance Test

| Structural role | Healthcare                                | Electricity                                     | Households                         | Invariant status                 |
|-----------------|-------------------------------------------|-------------------------------------------------|------------------------------------|----------------------------------|
| Q               | Service/workforce requirement             | Reliability-conditioned electricity requirement | Essential household requirement    | Stable                           |
| D               | Regular sustainable staffing              | Dependable deliverable capacity                 | Stable recurring resources         | Stable                           |
| C_S             | Normal renewable staffing flexibility     | Planned reserve/flexibility                     | Renewable household buffer         | Stable                           |
| C_E             | Extraordinary overtime/purchased staffing | Exceptional emergency substitution              | Exceptional gap-bridging coping    | Stable role, different mechanism |
| F               | Volume/access/continuity                  | Reliable electricity delivery                   | Essential social reproduction      | Stable                           |
| Restoration     | Durable staffing/access recovery          | Dependable capacity/flexibility addition        | D↑ or Q↓ with reduced coping       | Stable                           |
| Discontinuity   | Closures/waits/backlog                    | Curtailement/unserved energy                    | Food/housing/utility/car e failure | Stable                           |

The strongest combined result is that the logical positions remain stable even though their physical measures differ radically. This is the form of invariance expected from a structural grammar. The test does not require identical mechanisms; it requires stable relational roles.

## 7. Computed Evidence Matrix

To avoid a misleading single score, the synthesis computes support by proposition rather than assigning equal numerical weights to heterogeneous evidence. Each proposition is classified Supported, Partially Supported, Operationally Supported, Indeterminate, or Rejected.

| Proposition                                                         | Healthcare             | Electricity                                                   | Households                                       | Combined judgment                                |
|---------------------------------------------------------------------|------------------------|---------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|
| P1. Q and D can be separated                                        | Partial/Yes            | Yes                                                           | Yes, target-sensitive                            | Supported operationally                          |
| P2. C_S and C_E are conceptually separable                          | Partial                | Yes                                                           | Yes                                              | Supported operationally                          |
| P3. Compensation/flexibility can bridge mismatch                    | Observed descriptively | Operational possibility; standard science supports buffering  | Plausible; not directly observed longitudinally  | Partially supported                              |
| P4. Observable continuity can coexist with altered substrate/margin | Observed               | Supported structurally                                        | Supported only at aggregate/distributional level | Supported                                        |
| P5. Compensated Continuity as frozen SR configuration               | Partial direct support | Not yet directly confirmed                                    | Not yet directly confirmed                       | Partial support                                  |
| P6. Compensatory Inversion predicts later deterioration             | Indeterminate          | Indeterminate                                                 | Indeterminate                                    | Unvalidated                                      |
| P7. Restoration should reverse extraordinary dependence             | Prospective            | Strongly coherent with adequacy planning; prospective SR test | Prospective                                      | Operationally supported; predictive test pending |
| P8. Same grammar translates across heterogeneous domains            | Yes                    | Yes                                                           | Yes                                              | Supported                                        |
| P9. SR adds predictive value beyond established domain models       | Not tested             | Not tested                                                    | Not tested                                       | Unvalidated                                      |

## 8. Combined Status Calculation

The evidence can be summarized without pretending that ordinal categories are probabilities. Of the nine frozen cross-domain propositions above:

| Status                                 | Count | Meaning                                                        |
|----------------------------------------|-------|----------------------------------------------------------------|
| Supported / Supported operationally    | 4     | P1, P2, P4, P8                                                 |
| Partially supported                    | 2     | P3, P5                                                         |
| Operationally coherent but prospective | 1     | P7                                                             |
| Unvalidated / indeterminate            | 2     | P6, P9                                                         |
| Rejected at metatheory level           | 0     | No core relation has yet been falsified across the three tests |

This count should not be converted into a percentage of metatheory truth. The propositions have different scientific importance. P6 and P9—the predictive propositions—carry more weight for future validation than several operational propositions combined.

## 9. The Strongest Combined Finding

Across all three domains, the most defensible common proposition is:

Observable continuity does not by itself establish durable structural sufficiency. Healthcare demonstrates this directly through changing staffing composition under continued service. Electricity demonstrates why reliable output must be interpreted together with dependable margin and flexibility. Households demonstrate why aggregate continuity can conceal distributional exposure, while also showing that hidden compensation cannot simply be assumed.

The finding is diagnostic rather than predictive. It identifies a recurrent separation between what a system delivers and the structure carrying that delivery.

## 10. What the Three Papers Changed in SR

- Human Buffer is no longer treated as the universal mechanism; it is one domain-specific form of C.
- Normal resilience, reserve, slack, and flexibility are explicitly separated from extraordinary compensation.
- Debt is rejected as a universal household compensation proxy.
- Observable function is multidimensional; volume cannot erase access, latency, or distributional deterioration.
- Cross-domain similarity is defined by relational role rather than identical measurement units.
- Restoration is a co-equal branch of the grammar rather than an exception to a collapse trajectory.
- Compensatory Inversion remains prospective rather than being declared from suggestive descriptive patterns.

## 11. The Structural Grammar That Survived

Requirement ↔ Durable Capacity ↔ Sustainable Flexibility / Extraordinary  
Compensation ↔ Observable Function

This is now the empirically disciplined core of the Q–D–C–F side of SR. The grammar does not state that every positive gap produces compensation, that every compensation is depleting, or that every constrained buffer produces collapse. It states that these are separable positions whose relationships can be measured and whose transitions can be prospectively tested.

## 12. Cross-Domain Compensated Continuity

The combined evidence suggests three levels of support:

| Level                              | Definition                                                 | Current domain evidence             |
|------------------------------------|------------------------------------------------------------|-------------------------------------|
| Level 1 — Structural compatibility | The grammar can be operationalized without changing roles. | Healthcare, electricity, households |



|                                           |                                                              |                                         |
|-------------------------------------------|--------------------------------------------------------------|-----------------------------------------|
| Level 2 — Observed compensated continuity | C rises/changes under pressure while substantial F persists. | Healthcare strongest; others incomplete |
| Level 3 — Predictive compensation signal  | C state/trajectory improves prediction of future F.          | Not yet established in any domain       |

SR has crossed Level 1 and has partial Level 2 evidence. It has not crossed Level 3.

### 13. Why This Is Not Generic Confirmation by Analogy

A generic analogy could map almost any stressed system onto demand, capacity, buffer, and output. The present synthesis reduces that risk in four ways: the variable roles were frozen before the unified result; electricity rejected the naive equation of buffering with pathology; households rejected aggregate debt as compensation; and healthcare preserved the distinction between service volume and access. These constraints demonstrate that the grammar can lose interpretive claims even when the broad architecture survives.

Nevertheless, operational recurrence alone cannot establish unique SR value. Systems theory, resilience engineering, resource adequacy, household finance, operations research, and organizational theory already contain many of the component relations. SR's residual scientific burden therefore remains comparative.

### 14. Residual-Value Test

$$\text{SR\_residual value} = \text{performance}(\text{SR-augmented model}) - \text{performance}(\text{best domain baseline})$$

A positive residual requires more than statistical significance. The SR augmentation should improve out-of-sample forecast accuracy, calibration, early-warning lead time, or intervention prioritization. The decisive future test is therefore not whether Q, D, C, and F exist. It is whether their SR organization reveals something the strongest existing domain model misses.

### 15. Unified Statistical Architecture

$$M0\_d: F\_i(t+h) = f\_d(F\_i(t), Q\_i(t), D\_i(t), Z\_i(t))$$

$$M1\_d: M0\_d + C\_E(i,t) + \Delta C\_E(i,t) + [G\_C(i,t) \times \Delta C\_E(i,t)]$$

The subscript d denotes domain. Each domain retains its own estimator and physical units. Cross-domain comparison occurs through standardized predictive improvement and directional hypothesis performance, not by pooling raw units.

$$\Delta\text{Perf}_d = \text{Perf}(\text{M1}_d) - \text{Perf}(\text{M0}_d)$$

The central metatheoretical prediction is that  $\Delta\text{Perf}$  should be meaningfully positive in more than one domain and preferably in all domains where C\_E can be validly measured.

## 16. Cross-Domain Validation Threshold

| Criterion                                                 | Current status             |
|-----------------------------------------------------------|----------------------------|
| Three heterogeneous domains                               | Met                        |
| At least one engineered/non-human domain                  | Met                        |
| Frozen relational roles retained                          | Met                        |
| Invalid proxies rejected                                  | Met                        |
| At least one direct compensated-continuity case           | Partially met: healthcare  |
| At least two domains with incremental out-of-sample value | Not met                    |
| At least one prospective CI test completed                | Not met                    |
| At least one restoration branch observed and scored       | Not yet formally completed |
| Independent replication                                   | Not met                    |

The Structural Grammar therefore has preliminary cross-domain empirical support but does not yet satisfy the protocol's validation threshold.

## 17. What Would Falsify the Unified Grammar

- Valid Q–D gaps repeatedly fail to relate to any measurable buffering, substitution, or functional consequence across domains.
- Compensation variables add no predictive or diagnostic information beyond standard domain measures.

- Compensatory Inversion repeatedly fails to precede deterioration once the prospective windows close.
- Restoration does not reduce extraordinary compensation dependence or improve function when the measured gap closes.
- Cross-domain applications require changing the meaning of Q, D, C, or F after results are known.
- Independent researchers cannot reproduce the grammatical mapping without author-specific interpretation.

## **18. What Would Strengthen the Unified Grammar**

- Matched healthcare panels show compensation trajectory predicts later access or workforce deterioration beyond standard variables.
- Regional electricity panels show exceptional substitution contains predictive information beyond reserve margin, weather, outage, and adequacy models.
- Longitudinal household microdata demonstrate gap → identified coping → temporary continuity → restoration or later discontinuity.
- A restoration case is prospectively classified correctly before outcomes are known.
- The same directional compensation relation appears in a fourth hostile domain without changing the grammar.
- Independent replication reproduces both the mappings and the predictive increment.

## **19. Current Metatheory Judgment**

The combined evidence supports a careful conclusion. SR is empirically more than a collection of unrelated metaphors because the same relational roles can be operationalized across three heterogeneous systems while preserving explicit negative classifications. However, the evidence is not yet sufficient to conclude that SR has demonstrated unique predictive metatheoretical value.

Operational invariance: achieved

Direct cross-domain compensated-continuity evidence: partial

Predictive metatheoretical validation: not yet achieved

Accordingly, the current status is: developing diagnostic structural metatheory with preliminary cross-domain empirical support.

## **20. The Diagnostic Role After Three Tests**

The empirical series strengthens SR's diagnostic identity. The grammar is already useful for asking whether observed function is being produced by durable capacity, sustainable flexibility, extraordinary compensation, or some changing combination. This is a diagnostic decomposition. Prediction requires the additional step of showing that the decomposition forecasts future state transitions better than conventional indicators.

Diagnosis now; prediction only after calibration

This preserves SR's diagnostic role without claiming an evidentiary level it has not yet earned.

## **21. Unified Finding on Surface and Substrate**

The three papers converge on a single structural distinction:

Surface function and substrate condition are related but not identical variables.

Healthcare makes the distinction visible through staffing composition. Electricity makes it visible through reliability margin and the separation of dependable capacity from flexibility. Households make it visible through the difference between aggregate continuity and group-level essential-resource pressure. The exact substrate differs, but the analytical separation survives.

## **22. Unified Finding on Compensation**

Compensation cannot be defined by form alone. Overtime can be sustainable or extraordinary; reserve can be ordinary or exceptional; borrowing can be normal smoothing or distress. The correct classification depends on function, renewability, baseline, cost, persistence, and the presence of a measured gap.

Compensation status =  $f(\text{role, baseline, renewability, persistence, cost, gap})$

This is one of the most important refinements generated by combining the papers. It prevents SR from becoming a theory in which every buffer is interpreted as deterioration.

## **23. Unified Finding on Restoration**

The grammar is symmetric. A system can move out of the warning state by increasing D, reducing Q, replacing C\_E with C\_S, improving representation, or reorganizing the target. Successful restoration must therefore be scored as a prediction success for the recovery branch, not as a failure to observe collapse.

$$G\_C \downarrow + C\_E \text{ dependence } \downarrow + F \text{ maintained/improved} = \text{restoration}$$

This principle is especially important in electricity but applies equally to staffing and households.

## **24. The Next Empirical Step**

The domain-count phase should now stop. Adding more descriptive cases would provide diminishing scientific value. The next phase is predictive. The healthcare panel should be built first because it currently offers the most direct compensation measures. Electricity should follow with a harmonized regional event panel. Household validation should wait for suitable longitudinal microdata.

The next empirical papers should therefore test model performance, not merely structural fit.

## **25. Unified Canonical Empirical Statement — August 2026**

Across Canadian healthcare, North American electricity, and Canadian households, the SignalRupture Structural Grammar can be translated using stable relational roles for effective requirement, durable capacity, sustainable flexibility, extraordinary compensation, and observable function. The three domains do not exhibit identical mechanisms and do not provide equal evidentiary strength. Canadian healthcare supplies direct descriptive evidence that substantial observable function can coexist with increased reliance on compensatory staffing. Electricity demonstrates that the grammar survives a non-human engineered domain while requiring a strict distinction between normal

resilience and extraordinary substitution. Households demonstrate that the grammar survives a heterogeneous social domain only when compensation is identified from purpose and sequence rather than inferred from aggregate debt or hardship.

The combined evidence supports the diagnostic proposition that observable continuity is insufficient to establish durable structural sufficiency. It does not yet validate the stronger predictive proposition that the trajectory of extraordinary compensation reliably forecasts functional discontinuity across domains. That claim remains frozen for longitudinal and out-of-sample testing.

## **26. Conclusion**

The first three empirical papers substantially clarify what SR can and cannot currently claim. The Structural Grammar has survived translation across three heterogeneous domains without requiring identical physical mechanisms. It has also survived corrective pressure: electricity forced SR to distinguish resilience from extraordinary compensation, and households forced SR to reject aggregate debt as a shortcut for coping. These are not cosmetic refinements; they demonstrate that the grammar contains exclusion rules and can constrain interpretation.

The strongest current empirical result is not 'SR confirmed.' It is that the structural separation among requirement, durable capacity, compensation/flexibility, and observable function is measurable across multiple domains, and that Canadian healthcare already displays a descriptive compensated-continuity configuration. The strongest unresolved question is whether compensation state provides incremental prospective information beyond established domain science.

SR therefore enters its next phase with a narrower but stronger status: a developing diagnostic structural metatheory with preliminary cross-domain empirical support. Its metatheoretical future depends on prediction. If the frozen SR augmentation improves out-of-sample forecasts across multiple domains and survives independent replication, the Structural Grammar will have demonstrated residual empirical value. If it does not, the architecture must narrow or converge with the established fields from which its components are drawn.

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# Appendix A. Unified Scorecard

| Criterion                | Healthcare | Electricity | Households       | Unified status       |
|--------------------------|------------|-------------|------------------|----------------------|
| Frozen mapping retained  | Yes        | Yes         | Yes              | Passed               |
| Q–D relation operational | Partial    | Yes         | Yes              | Passed operationally |
| C_S/C_E separation       | Partial    | Yes         | Yes conceptually | Passed operationally |

|                                 |                |             |         |                            |
|---------------------------------|----------------|-------------|---------|----------------------------|
| Observed Compensated Continuity | Partial/direct | Not yet     | Not yet | Partial                    |
| CI observed                     | No             | No          | No      | Indeterminate              |
| Restoration branch test         | Pending        | Prospective | Pending | Pending                    |
| Incremental prediction          | Pending        | Pending     | Pending | Not established            |
| Domain translation              | Passed         | Passed      | Passed  | Three-domain threshold met |

## Appendix B. Unified Research Queue

| Priority | Study                                          | Purpose                                                                                    |
|----------|------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1        | Canadian healthcare matched longitudinal panel | Test whether C_E and $\Delta C_E$ improve prediction of access/workforce outcomes.         |
| 2        | Regional electricity event panel               | Test exceptional-substitution increment beyond mature adequacy models.                     |
| 3        | Longitudinal household microdata study         | Identify gap $\rightarrow$ coping $\rightarrow$ continuity $\rightarrow$ outcome sequence. |
| 4        | Representation-Rupture empirical paper         | Test X-R side of the Structural Grammar separately.                                        |
| 5        | Independent replication package                | Test whether non-author researchers reproduce mappings and results.                        |